WORKPULSE: EMPLOYEE ATTENDANCE AND WORK ANALYTICS INFORMATION SYSTEM FOR MVL

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BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

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Chapter 1

INTRODUCTION

Background of the Study

Employee attendance and work information management is a vital component of organizational operations, as it directly affects productivity, accountability, and decision-making. Globally, organizations are increasingly adopting digital and analytics-driven human resource systems to replace manual and outdated attendance methods that often result in errors, inefficiencies, and limited access to actionable insights. Recent literature highlights that the integration of biometric-based system is revolutionizing human resource management by enhancing operational efficiency, minimizing labor-related inefficiencies, and eliminating the administrative burdens of traditional tracking methods (ljiga et al., 2025). Modern workforce analytics systems enable organizations to record attendance accurately, monitor work patterns, and generate data-driven reports that support effective human resource management.

Despite these global advancements, studies reveal that many organizations still rely on disconnected or partly manual processes to manage employee information, attendance, leave, and payroll. These practices often result in inconsistent records, delayed approvals, limited employee transparency, inefficient reporting, and weak accountability. Basic systems may also lack appropriate authentication, authorization, audit logging, and protection for sensitive employee information, which can compromise employee trust and organizational credibility. Furthermore, traditional payroll and security systems such as manual logs or PIN codes contribute heavily to human error and unauthorized entry, explicitly highlighting the critical need for biometric verification to reduce proxy attendance practices like "buddy punching" (International Journal of Science and Research Archive, 2026).

In the Philippine context, organizations are rapidly transitioning toward digital transformation in response to the growing demand for efficient, transparent, and technology-driven workplace systems. The Philippine HR technology market is experiencing significant growth as employers increasingly rely on cloud-based platforms and digital portals for faster, more efficient workforce processing (IMARC Group, 2025). This push for digitalization is evident even at the national level, with recent milestone initiatives like the Philippine Civil Service Modernization Project launched in late 2025, which seeks to harness digital tools to automate human resource and payroll systems for better efficiency and transparency (Civil Service Commission, 2025).

However, many private and local organizations continue to face challenges such as fragmented attendance records, inefficient monitoring of work hours, and limited analytical capability. A centralized and secure workforce information management system is therefore needed to improve record management, administrative workflows, employee access, and data-supported decision- making.

In response to these challenges, this study proposes WorkPulse, a secure web-based workforce information management system with integrated fingerprint-based attendance verification. The project aims to centralize employee, attendance, leave, and payroll records while providing strict role-based access organized into administrator and employee access groups. By utilizing a fingerprint scanner to verify employee identity during time-in and time-out, the system aims to associate biometric attendance records with the verified employee, improving identity assurance and operational integrity. Data is securely managed using a cloud-hosted MongoDB Atlas platform to support centralized, authorized data storage and access. This study aligns with Sustainable Development Goal (SDG) 8: Decent Work and Economic Growth by promoting fair labor practices, and SDG 9: Industry, Innovation, and Infrastructure by integrating digital analytics and secure hardware solutions into organizational process.

Statement of the Problem

WorkPulse addresses employee information, attendance, leave, cash advance, and payroll processes that may be difficult to reconcile when maintained separately. Requirements were gathered through interviews. Usability was evaluated using the System Usability Scale (SUS), while functional behavior was examined through documented software tests. No separate workplace-challenges or system-quality survey was administered.

1. What workplace requirements and concerns were identified through interviews about attendance, employee information, leave, cash advances, and payroll?

2. How can WorkPulse centralize these records while providing administrator services and employee access to authorized personal information?

3. How can authorized manual attendance, corrections, notifications, and limited offline saving support the attendance workflow?

4. What level of perceived usability did WorkPulse obtain using the System Usability Scale?

5. How did the implemented attendance, payroll, authorization, and offline workflows perform in the documented functional tests?

Objectives of the Study

The primary objective is to design, develop, and evaluate WorkPulse, a web-based employee attendance and work analytics information system for MVL. The system centralizes workforce records, provides administrator and employee services, and supports fingerprint attendance at the configured local workstation together with authorized manual attendance procedures.

1. To identify workplace requirements through interviews about employee information, attendance, leave, cash advances, and payroll.

2. To centralize employee, attendance, leave, cash advance, and payroll records using MongoDB Atlas.

3. To implement administrator and employee access groups with record ownership controls and Regular, Extra, and Manager / Supervisor employee classifications.

4. To provide an employee portal for authorized personal records, attendance, leave and cash advance requests, payroll summaries, and notifications.

5. To implement authorized attendance corrections with required reasons, correction history, employee notification, and protection of attendance included in settled payroll.

6. To generate descriptive workforce dashboards and reports and provide separately identified analytical insights for administrative review.

7. To implement limited offline saving and later synchronization of designated attendance actions, leave requests, and cash advance requests.

8. To evaluate perceived usability using SUS and verify selected business rules and error handling through documented software tests.

9. To integrate fingerprint-based time-in and time-out with the configured local workstation and document its operating requirements and evaluation limitations.

Scope and Delimitation

WorkPulse provides administrator and employee access groups. Employees are classified as Regular, Extra, or Manager / Supervisor. Manager and supervisor represent one classification and share one company hourly-rate setting. These classifications do not create a separate management portal or confer administrator authority.

The system covers employee records, fingerprint attendance at a configured local kiosk, authorized administrator manual attendance, eligible attendance corrections, leave requests, cash advance workflows, payroll processing, personal payslip summaries, notifications, company settings, and audit logging. Attendance corrections retain a reason, administrator identity, timestamp, and history. Server validation protects records included in paid, settled, or carried payroll.

Cash advance management distinguishes request approval from confirmation that cash was given. Confirmed disbursements establish outstanding balances; recorded repayments reduce those balances. Carried payroll balances are unpaid employee earnings transferred between payroll statements and are distinct from outstanding cash advance debt. The system records settlement; it is not integrated with a bank or payment gateway.

Descriptive dashboards summarize stored workforce events and totals. The application also includes statistical forecasting, attendance-risk indicators, and arrival anomaly analysis for administrative review. Their presence does not establish measured forecasting accuracy, and their outputs do not directly measure individual productivity or automatically determine disciplinary actions.

Limited offline operation uses a prepared browser and a valid existing session. Eligible manual attendance/corrections, leave requests, and cash advance requests may be queued locally, and selected previously cached information may be viewed. Employee requests synchronize on reconnection after session validation; attendance synchronization requires administrator password confirmation. All synchronized actions remain subject to current server rules.

Fingerprint capture and matching require a compatible reader, locally installed components, and a Windows workstation running the local matching workflow. The scanner is not directly attached to the remote Render backend. Fingerprint attendance, fresh login, approvals, payroll settlement, and cash-given confirmation require their online services. Browser storage may fail or be cleared, and cached information may be outdated.

Research data gathering consisted of interviews for requirements and SUS for usability. No separate questionnaire produced workplace-problem frequencies or security, performance-efficiency, and reliability ratings. Functional results describe the tested local application and controlled browser environment. Physical fingerprint accuracy, live deployment recovery, and forecasting accuracy are not established by those software tests.

Significance of the Study

Administrators and Management. Centralized records, attendance history, request processing, and payroll summaries support review and reduce repeated record reconciliation. Traceable corrections provide accountability when manual adjustments are required.

Employees. The employee portal provides access to authorized attendance, leave, cash advance, and payroll information. Notifications improve visibility of changes affecting an employee's records.

IT Practitioners and Developers. The project demonstrates browser interfaces, cloud record storage, hardware-dependent attendance integration, server authorization, and limited offline synchronization within one application.

Future Researchers. WorkPulse provides a basis for further usability, biometric, outage-recovery, and analytical evaluation using documented methods and measured outcomes.

Definition of Terms

Attendance Correction. An authorized modification of eligible attendance, retained with its reason, administrator identity, timestamp, and history.

Attendance Monitoring. Viewing attendance dates, statuses, and recorded work sessions.

Audit Logging. Recording significant administrative and security activities for review and accountability.

Authentication. Verifying an account before permitting access to WorkPulse.

Authorization. Restricting operations and records according to authenticated permissions and ownership.

Biometric Identification. Comparing a captured fingerprint against enrolled reference templates to identify an employee in the local attendance workflow.

Carried Payroll Balance. Unpaid employee earnings transferred from an earlier payroll period into a subsequent payroll statement.

Cash Advance. An employee request for an advance, subject to review, cash-given confirmation, outstanding-balance tracking, and repayment.

Cloud Database. Remote record storage used by WorkPulse through MongoDB Atlas.

Dashboard. A presentation of workforce indicators, tables, and charts.

Data Validation. Checking input and synchronized actions against required formats and business rules.

Descriptive Analytics. Summarizing recorded information without presenting the summary itself as a forecast.

Employee. An authenticated member of the employee access group, classified as Regular, Extra, or Manager / Supervisor.

Employee Notification. A system-generated notice about relevant recorded events, administrative changes, or request decisions.

Employee Portal. The employee interface for personal workforce records, requests, summaries, and notifications.

Encryption. Transforming data to protect it against unauthorized disclosure.

Fingerprint Scanner. The supported local reader that captures fingerprint samples.

Fingerprint-Based Attendance. Recording time-in or time-out following identification using the configured reader and matching components.

Leave Management. Submission, review, approval or rejection, and recording of employee leave requests.

Manager / Supervisor. One employee classification, alongside Regular and Extra, that does not automatically confer administrator access.

MongoDB Atlas. The cloud-hosted database platform used for WorkPulse workforce records.

Offline Queue. Account-specific local storage of supported actions awaiting server acceptance.

One-Time Password. A temporary email verification code used in an applicable authentication step.

Outstanding Cash Advance Balance. Confirmed cash advances remaining unpaid after recorded repayments.

Payroll Management. Preparation, review, carry-forward, and settlement recording of employee earnings and deductions.

Payslip Summary. A display of current earnings, deductions, carried earnings, and the resulting payout.

Role-Based Access Control. Server-side restrictions based on authenticated account permissions and employee-record ownership.

Session Management. Creating, validating, expiring, and terminating authenticated access.

Synchronization. Submitting locally queued actions for server validation and reconciliation after connectivity returns.

System Usability Scale. The ten-item questionnaire used in this study to evaluate perceived usability on a score from 0 to 100.

Time-In and Time-Out. Recorded beginning and ending work times, entered through supported fingerprint attendance or authorized manual attendance.

WorkPulse. The employee attendance and work analytics information system developed for MVL.

Chapter 2

METHODOLOGY

This chapter describes the development process, architecture, data gathering, usability scoring, software verification, storage, and ethical considerations of WorkPulse. Requirements were gathered through interviews. SUS was the only respondent questionnaire used; no separate workplace-challenges or system-quality questionnaire was administered.

Development Methodology

The development approach follows the Agile model through Scrum. Work is organized into a product backlog containing application requirements, defects, and improvements. Planning, design, implementation, testing, and review support successive revisions of the employee, attendance, leave, cash advance, payroll, and reporting modules.

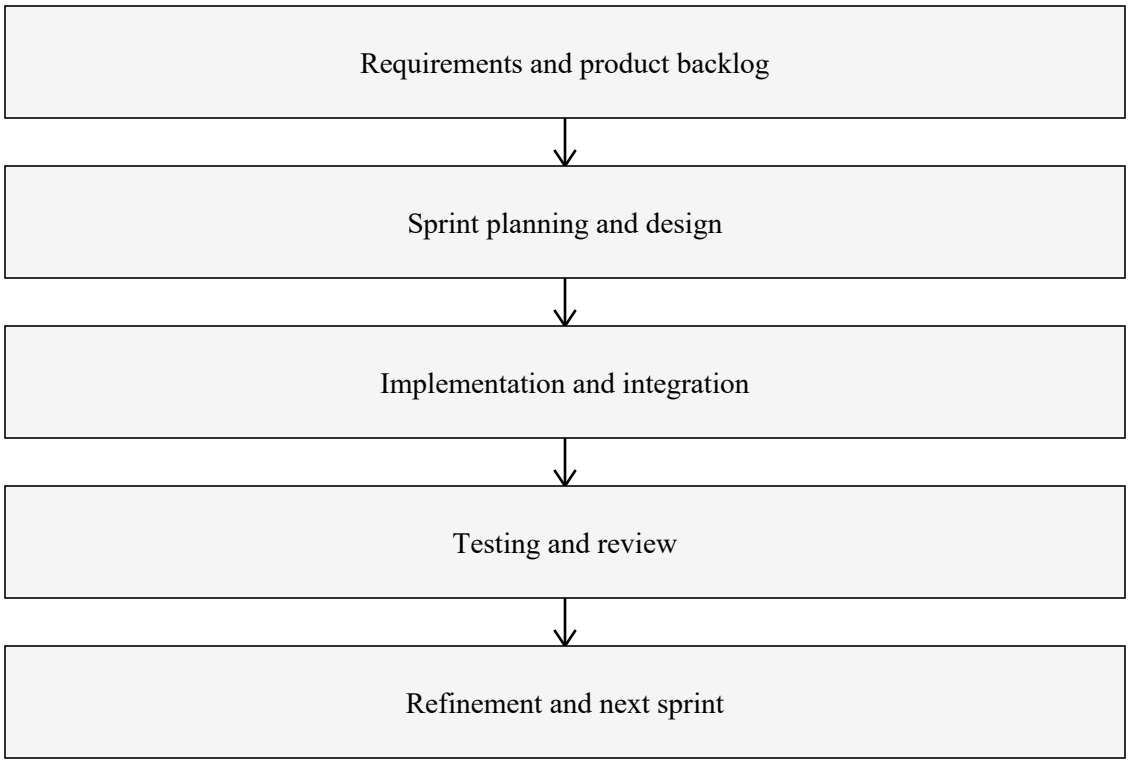


Figure 1. Iterative development workflow

Planning. Interview findings informed requirements for attendance, employee records, leave, cash advances, payroll, and employee access. Functional requirements include traceable

manual corrections, notifications, protected payroll settlement, and limited offline saving.

Design. The employee-management module maintains profiles and classifications. Payroll uses employee identifiers to prepare earnings, cash advance deductions, carried unpaid earnings, and summaries. Authorization separates administrator actions from employee self-service and restricts personal records to their owners.

Development. React and TypeScript provide browser interfaces. Node.js and Express implement business rules and protected routes. MongoDB Atlas stores cloud records. IndexedDB and a service worker support selected local storage and cached resources.

Testing. Backend tests examine business rules against controlled records, including invalid times, overlapping sessions, settled payroll restrictions, request ownership, notifications, and synchronization. Browser checks exercise actual IndexedDB and service worker behavior with simulated API responses. The frontend production build checks that the application can be compiled for release.

Deployment. The frontend is hosted on Vercel, the cloud Express backend on Render, and authoritative records in MongoDB Atlas. Fingerprint attendance remains dependent on the configured local Windows workstation. The test results reported in Chapter 3 apply to the local application and controlled browser environment rather than a complete live deployment evaluation.

System Architecture

The cloud application follows presentation, application, and data layers. Browser-local storage supplements these layers for supported offline operations. Fingerprint matching follows a separate hardware-dependent local execution path.

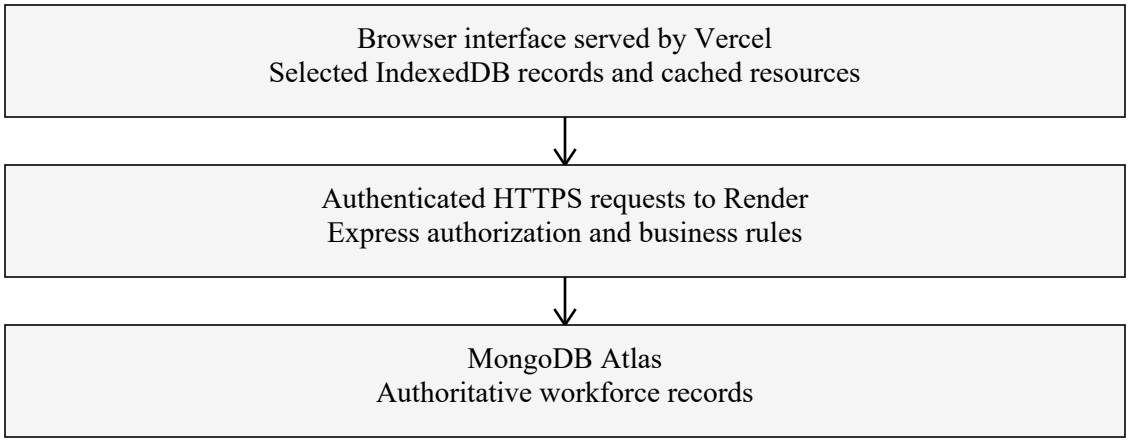


Figure 2. Cloud application and local attendance processing

Presentation Layer. React provides administrator, employee, and kiosk interfaces. Browser interface visibility is supported by server-side authorization rather than treated as an access control by itself.

Application Layer. Express routes handle authenticated requests, attendance rules, employee ownership, leave and cash advance workflows, payroll, notifications, audit logging, and duplicate-resistant synchronization.

Data Layer. MongoDB Atlas stores employee, account, attendance, leave, cash advance, payroll, settings, biometric, session, notification, and audit records. Locally queued actions are not authoritative cloud records until accepted by the server.

Local Fingerprint Processing. The browser captures samples through DigitalPersona reader components. The backend matching code launches a Windows PowerShell helper that loads the locally installed FingerJet runtime. This processing runs on the Windows machine hosting that matching backend, rather than directly on the remote Render service. A local backend using the cloud database still requires internet access for database operations. No completed cloud-to-local matching bridge is claimed in this study.

Email Integration. The backend supports configured email delivery for authentication and supported messages. The code includes SMTP through Nodemailer and Brevo HTTPS delivery. The research does not establish that both providers were used in production.

Tools and Technologies

React 19, TypeScript, Vite, and Tailwind CSS provide browser interfaces, typed development, production bundling, and styling. Recharts presents graphical workforce summaries.

Node.js and Express provide the backend runtime and protected application services. MongoDB Atlas and MongoDB/Mongoose operations support cloud record storage.

IndexedDB stores account-specific supported requests and selected cached data. A service worker caches application resources for prepared-device offline access.

DigitalPersona components and HID FingerJet support the local reader and matching workflow. The current helper depends on Windows PowerShell and the locally installed native runtime.

The implementation uses scrypt password derivation and AES-256-GCM protection for stored biometric payloads. Configuration remains necessary for operational encryption and authentication.

Vercel and Render host the frontend and cloud backend. Dotenv supports configuration, while Git supports source-code revision tracking.

Population and Locale of the Study

The revised study population comprises 30 MVL participants: two owners, two managers/supervisors, ten regular employees, and sixteen extra employees. These organizational categories are distinct from login access groups. Owners may use administrator access, while the other classifications belong to the employee access group. The listed population is not a claim that all participants were interviewed.

The locale is the MVL workplace in Salay Riles, Mangaldan, Pangasinan, Region I.

Table 1. Study population

Classification	Number
Owner	2
Manager / Supervisor	2
Regular employee	10
Extra employee	16

Classification	Number
Total	30

Data Gathering Techniques

Interviews. Interviews provided workplace requirements and concerns about attendance recording, employee information, leave, cash advances, and payroll. Interview findings are reported qualitatively. No survey frequencies, percentages, or direct quotations are assigned to these findings without corresponding records.

System Usability Scale. SUS was administered to evaluate perceived usability. It contains ten statements answered on a five-point response scale. The researchers' computed overall result was 80.83. The SUS questionnaire is distinct from a workplace-challenges survey or a separate questionnaire rating security, performance, and reliability; those additional questionnaires were not administered.

No separate formal observation study, expert-validation procedure, pilot questionnaire study, or Cronbach's alpha calculation is reported as completed in this methodology.

Sources and Treatment of Data

Primary research sources consist of interviews and SUS responses. Project source code, supplied interface illustrations, and recorded software test outcomes support the description and functional evaluation of WorkPulse. Existing technical literature provides background and context.

Interview findings are organized by workplace topic without converting them into uncollected survey statistics. SUS scores represent participant perceptions of usability. Software test records identify input conditions, expected behavior, actual outcomes, and pass/fail status in the specified environment. The population table describes participant categories; it does not provide interview response frequencies.

SUS Scoring

For each participant, odd-numbered SUS items contribute the response value minus one, and even-numbered items contribute five minus the response value. The ten contributions are added and multiplied by 2.5 to produce a participant score from 0 to 100. The overall SUS score is the mean of participant scores. This manuscript preserves the researchers' computed result of 80.83; it was not replaced or recalculated during the chapter revision. A SUS score is not a percentage of participants who approved the system.

Functional Verification Procedure

Local verification was performed on October 11, 2026. The backend suite exercised business rules using simulated database records and transaction behavior. The browser smoke test used an isolated Microsoft Edge profile, actual IndexedDB and service worker caching, and simulated server responses. Tests included valid and rejected attendance changes, payroll repayment behavior, account isolation, local saving, offline reload, reconnect synchronization, duplicate retries, and storage failure.

The 175 backend checks are software test cases, not respondent trials. Browser checks establish the observed behavior in the controlled environment. They do not constitute live Render/Vercel recovery tests, physical fingerprint performance measurements, or exhaustive infrastructure-failure testing. The expected and actual outcomes are presented in Chapter 3.

Limited Offline Operation

Offline use requires prior online sign-in, a valid existing session, and a prepared browser/device with the necessary cached resources. Eligible attendance actions, leave requests, and cash advance requests are retained in IndexedDB. Cached information may be stale. Pending indicates local retention; Needs Review indicates an unresolved validation issue; Synced indicates acceptance by the server.

Employee requests may synchronize on reconnection after session validation. Attendance synchronization requires administrator password confirmation. The server checks ownership, dates, record versions, payroll restrictions, and operation identifiers. Fresh login, fingerprint transactions, approvals, and settlement actions remain dependent on online services. Passwords and raw fingerprint samples are not stored in the offline queue.

Data Storage, Backup, and Recovery

MongoDB Atlas stores authoritative records under application authentication, ownership, validation, and authorization controls. Browser-local storage holds only selected cached records and supported queued actions. Stable identifiers prevent duplicate processing after an uncertain network response. Backup-format compatibility was exercised by software tests; those checks do not establish a successful restoration of the deployed production database. Browser storage may be cleared or unavailable and is not a replacement for database backup.

Biometric Evaluation Boundary

The application includes a configured fingerprint attendance workflow. The available evidence does not include completed physical-reader trial counts, failed-capture measurements, false identification outcomes, or measured processing times. Consequently, no numerical biometric accuracy or speed results are reported. Future biometric evaluation should use consenting participants, documented reader conditions, genuine and unrecognized-finger trials, and recorded timing measurements.

Ethical Considerations

WorkPulse handles personal, attendance, payroll, cash advance, and fingerprint information. Data handling is guided by transparency, legitimate purpose, proportionality, and restricted access. Research records should be anonymized where appropriate, and participation in evaluation should be voluntary. Statistical indicators support human review and do not automatically impose disciplinary or payroll decisions.

Manual attendance provides an alternative when fingerprint capture is unavailable. Corrections retain reasons and history and notify the affected employee when accepted. If the workstation itself is unavailable, a supervisor-verified paper log is a recommended contingency for later authorized entry. Organizational approval of that paper procedure is not asserted as a completed research result.

Chapter 3

DISCUSSION OF FINDINGS

This chapter presents workplace topics identified through interviews, the implemented WorkPulse modules, the researchers' SUS usability result, and verified software test outcomes. Interview findings, participant usability ratings, and functional tests are treated as separate forms of evidence.

Interview Findings and System Requirements

The interviews identified difficulty preparing payroll because amounts were manually entered in Excel for each employee. The owner also reported observations indicating a risk of unauthorized or proxy attendance, including the possibility of a time-in record without the employee being physically present. Other topics were monitoring work hours, transparency of employee records, and occasional inaccuracies in the attendance method used. These findings and reported observations guided the system requirements. No separate workplace-challenges questionnaire was administered, so they are presented qualitatively rather than as respondent frequencies or percentages. The reported attendance risk does not establish a verified incident of fraud.

Table 2. Interview topics and corresponding system responses

Interview topic	WorkPulse response
Difficulty preparing payroll through manual Excel entry per employee	Attendance-linked payroll preparation using recorded work hours, configured rates, repayment deductions, carried earnings, and payslip summaries
Owner-observed risk of unauthorized or proxy attendance	Configured fingerprint identification at time-in/time-out and administrator-controlled manual alternatives with traceable records
Monitoring employee work hours	Daily/date-range attendance views and recorded work sessions
Transparency of attendance and payroll records	Personal employee portal for authorized attendance and payroll summaries, request status, and relevant notifications
Occasional inaccuracies in the attendance method used	Recorded sessions, source indicators, validation, and authorized corrections with reasons and history

Implemented Features of WorkPulse

The following descriptions refer to the implemented application. Supplied interface illustrations are retained where useful, with corrected captions. They depict the version shown in the uploaded manuscript and are not evidence that every current change has already been deployed. The narrative describes the combined Manager / Supervisor classification and the current offline and payroll behavior.

Employee Information Management

Administrators maintain employee profiles and account access. Classifications are Regular, Extra, and Manager / Supervisor. Older supervisor records remain compatible with the combined classification; newly edited or created records use the unified management classification.

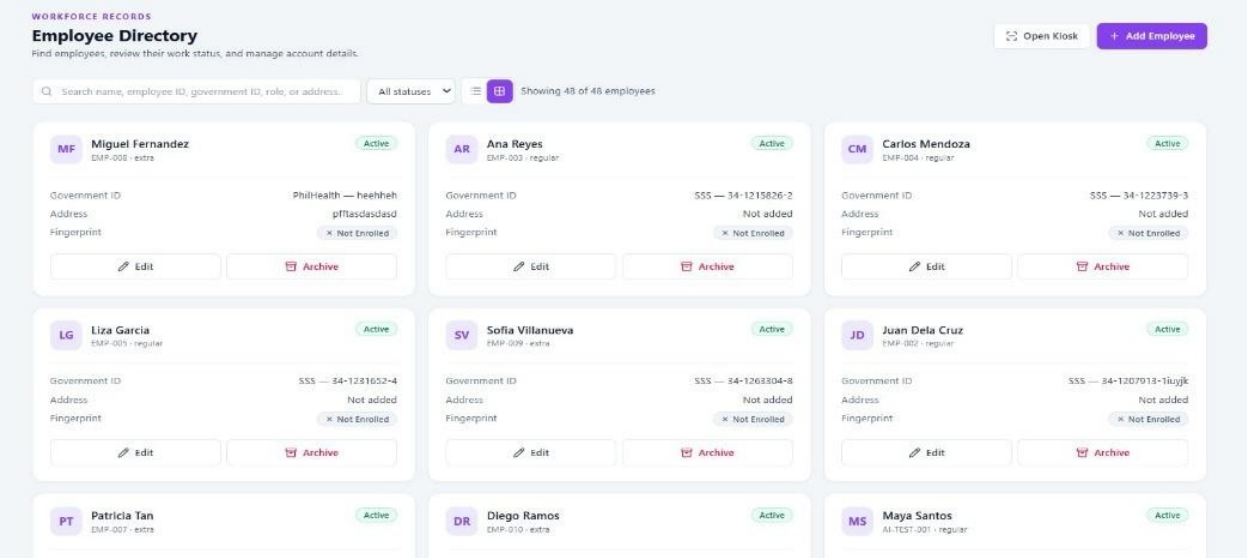


Figure 3. Employee directory interface supplied in the manuscript

Fingerprint-Based Attendance

The kiosk records time-in and time-out through the configured local reader and matching environment. Its current matching helper requires the local Windows backend. Physical recognition accuracy is not inferred from the application screenshots or software tests.

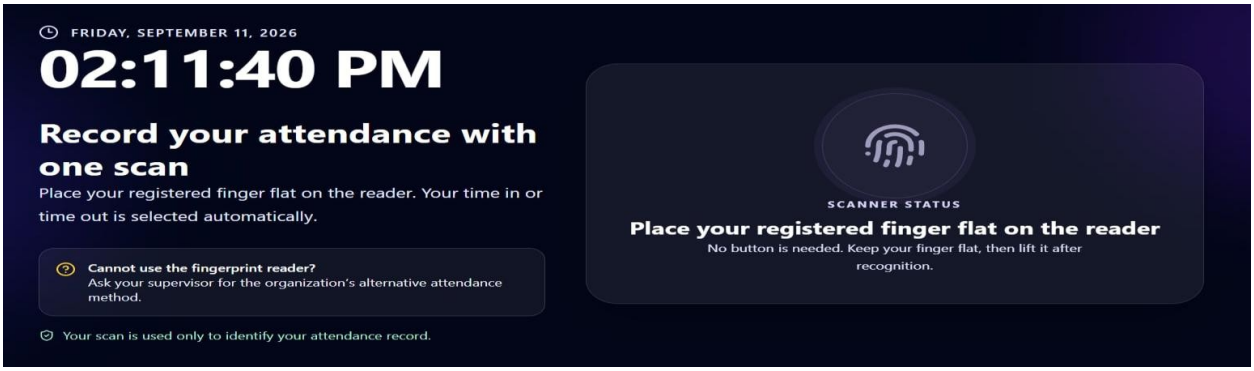


Figure 4. Fingerprint kiosk interface supplied in the manuscript

Attendance Monitoring and Corrections

Administrators review daily and date-range attendance and up to three sessions per employee. Eligible manual entries and corrections retain their source, reason, administrator, timestamp, and history. The affected employee receives a notification. Future times, overlapping sessions, stale records, and prohibited changes to settled payroll are rejected.

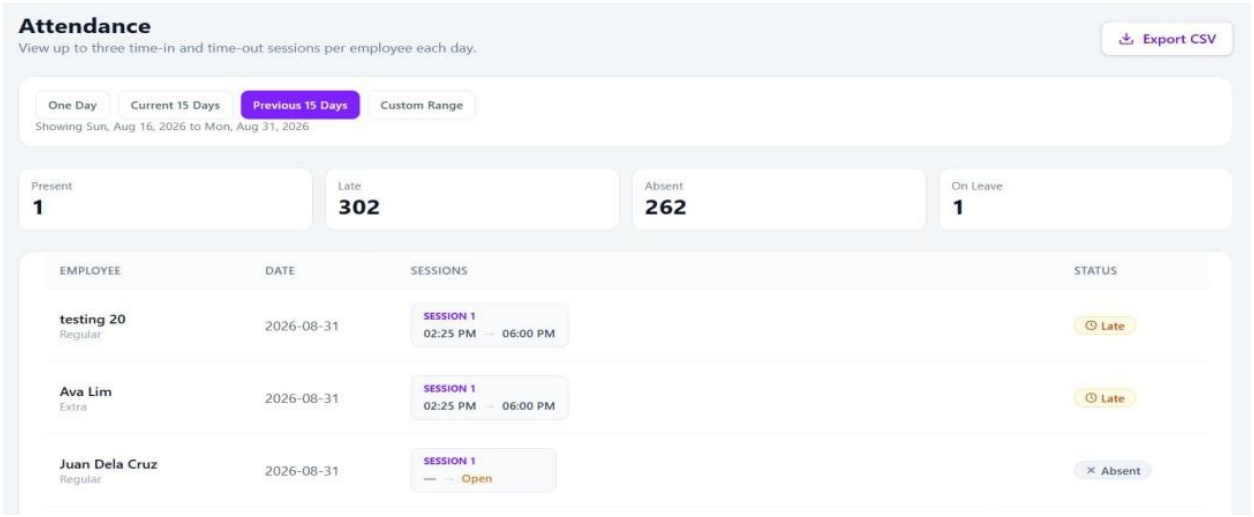


Figure 5. Attendance interface supplied in the manuscript

Employee Self-Service Portal

Employees view authorized attendance, leave, cash advance, and payroll records and receive relevant notifications. Source indicators distinguish manual and edited attendance. Selected cached views and supported queued requests remain available on a prepared browser during an outage.

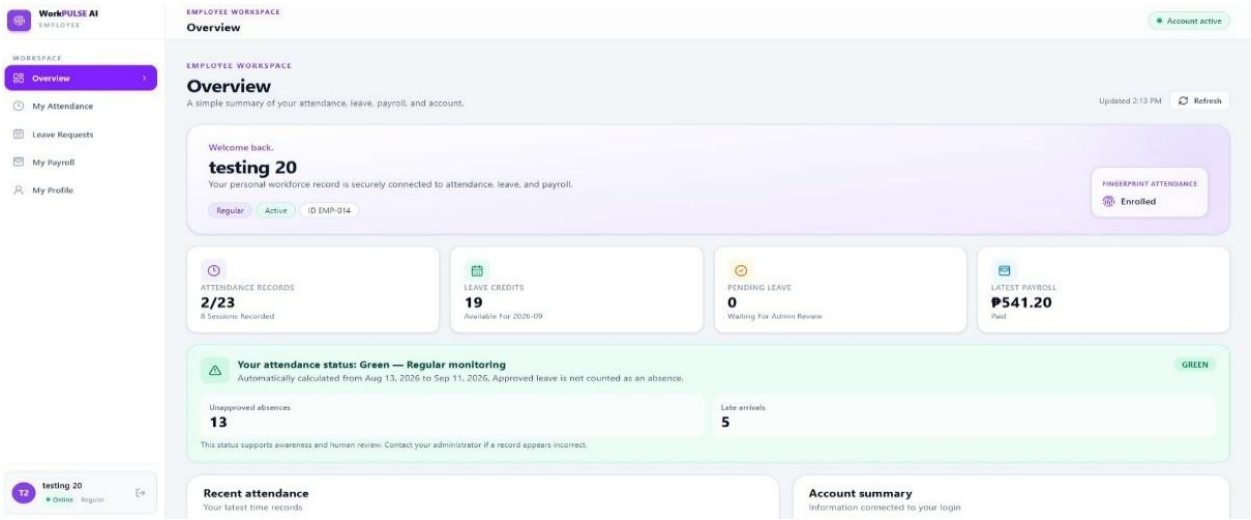


Figure 6. Employee overview interface supplied in the manuscript

Leave Management

Employees submit requests, and administrators review and process them. Pending requests with dates that have passed are distinguished from unresolved current or future dates. Offline saving does not approve a request; server acceptance and administrative decisions remain separate.

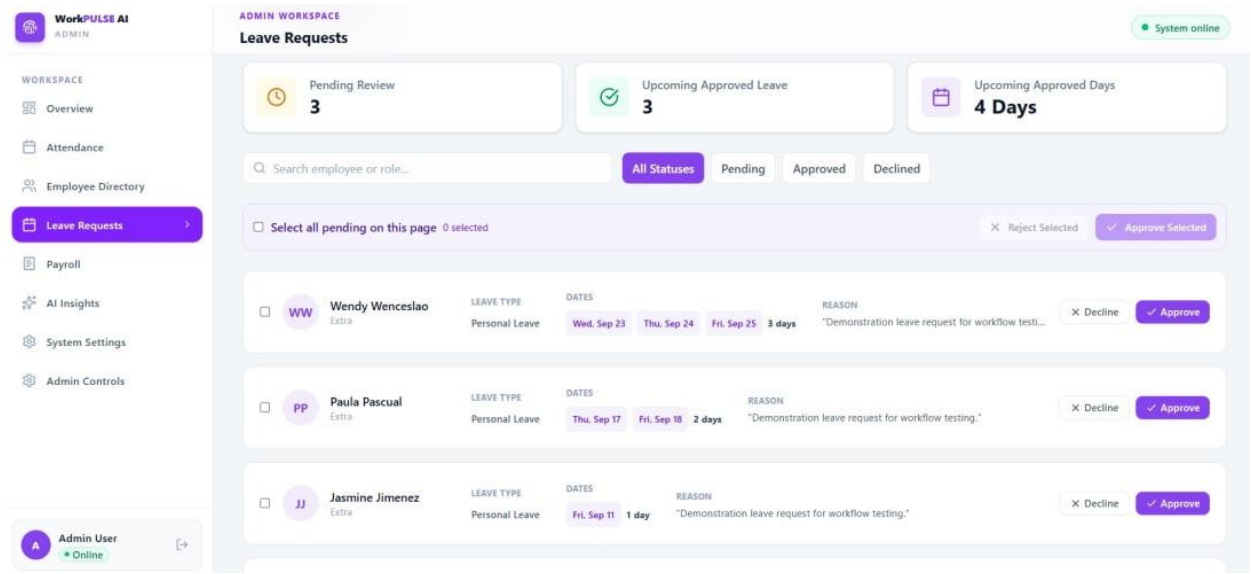


Figure 7. Leave-request interface supplied in the manuscript

Payroll Management

Payroll displays prepared earnings, unpaid amounts carried forward, deductions, and settlement status. Ready to Pay identifies a prepared amount, Carried to Next Period identifies unpaid earnings transferred forward, and Paid identifies confirmed settlement. These statuses reflect different stages of the same workflow.

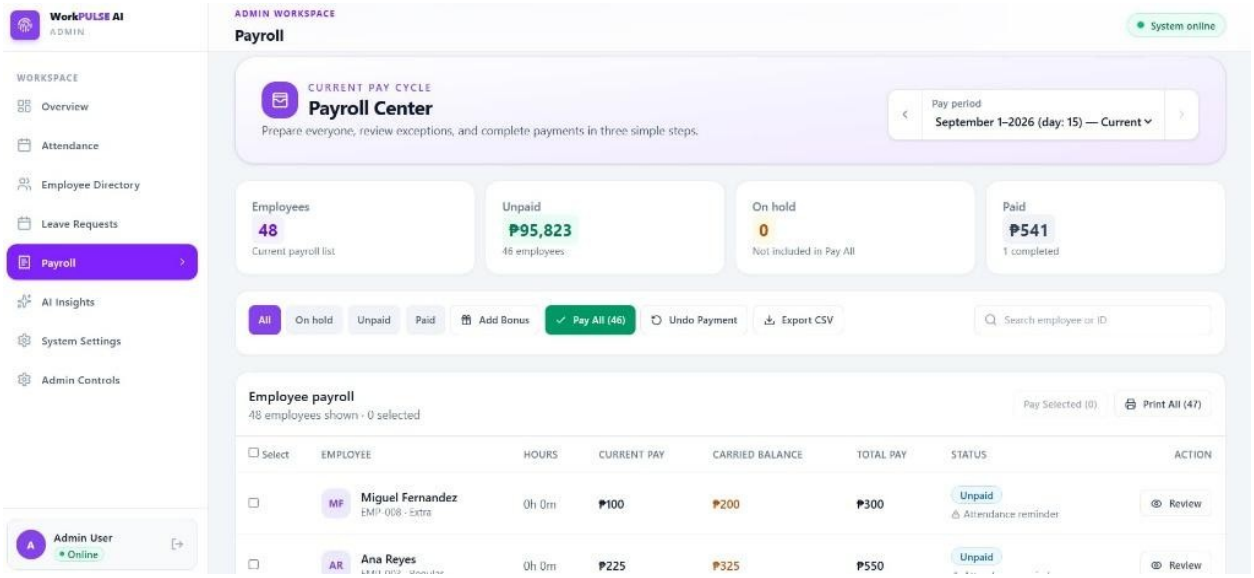


Figure 8. Payroll interface supplied in the manuscript

Payroll Summary and Cash Advance Repayment

Payslip summaries distinguish current earnings, carried unpaid earnings, repayment deductions, and total payout. Request approval alone creates no outstanding advance. Confirmed cash-given records create debt, and posted repayments reduce it. Eligible installments from included earnings periods accumulate when unpaid periods are settled together.

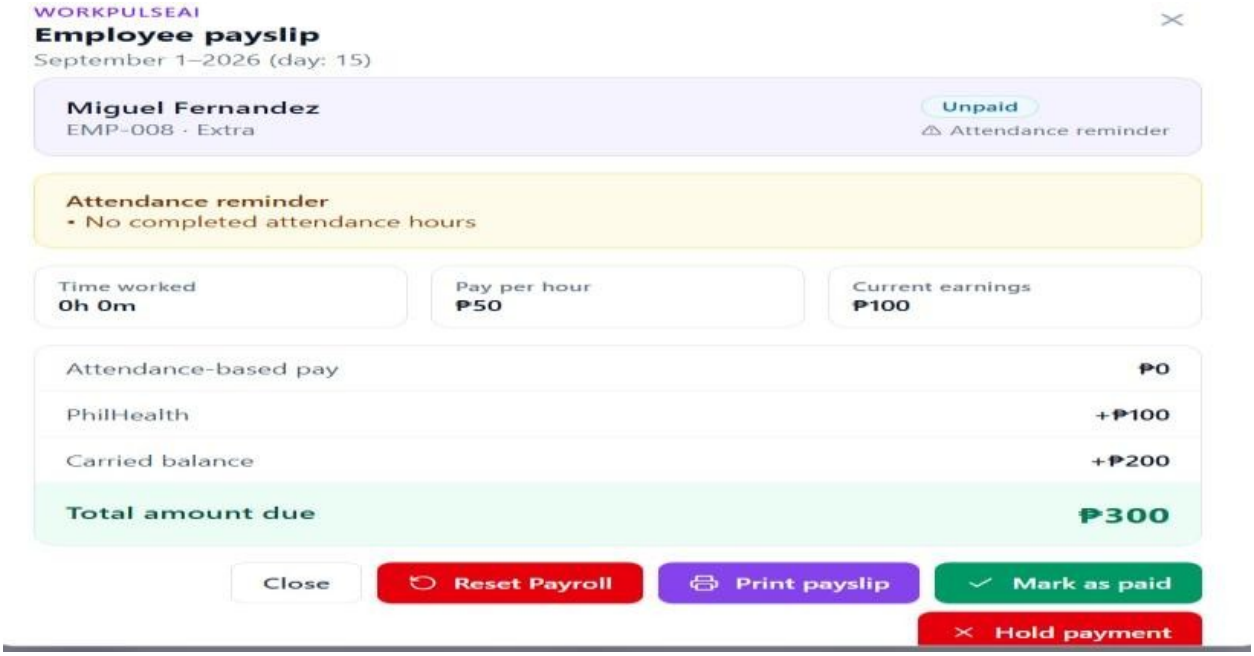


Figure 9. Payslip interface supplied in the manuscript

Workforce Reporting and Analytical Insights

Dashboards summarize workforce information. Statistical attendance forecasting, risk indicators, and arrival anomaly analysis support review. No measured forecasting-accuracy result is established by the functional checks reported here.

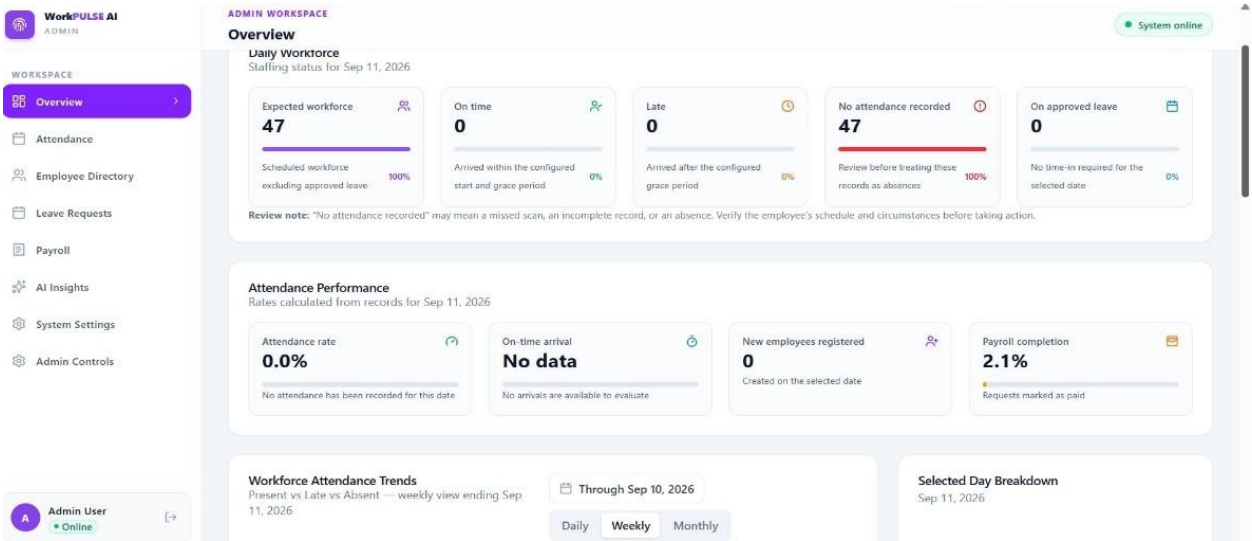


Figure 10. Workforce overview supplied in the manuscript

Authentication and Record Ownership

Password and configured email verification support access. Server-side authorization separates administrator operations from employee self-service and restricts employee records by authenticated ownership. Regular, Extra, and Manager / Supervisor classifications do not confer administrator privileges.

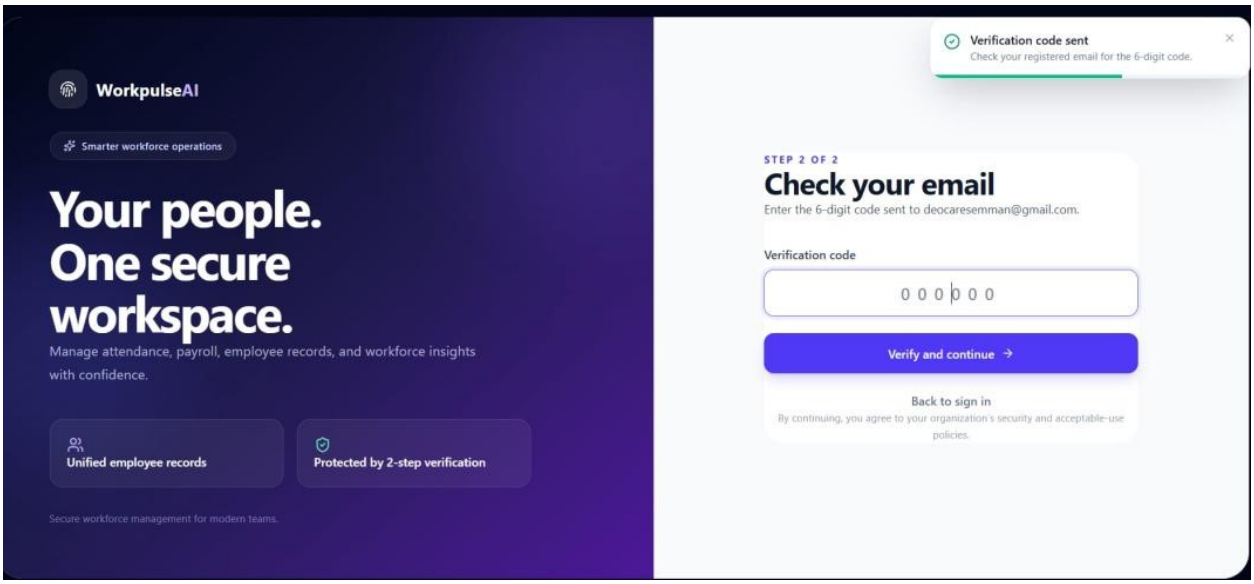


Figure 11. Email verification interface supplied in the manuscript

Cloud Storage and Audit Accountability

MongoDB Atlas stores authorized cloud records. Audit events and attendance history support review of relevant actions. Browser-local storage is limited to selected cached information and queued submissions rather than a full database replica.

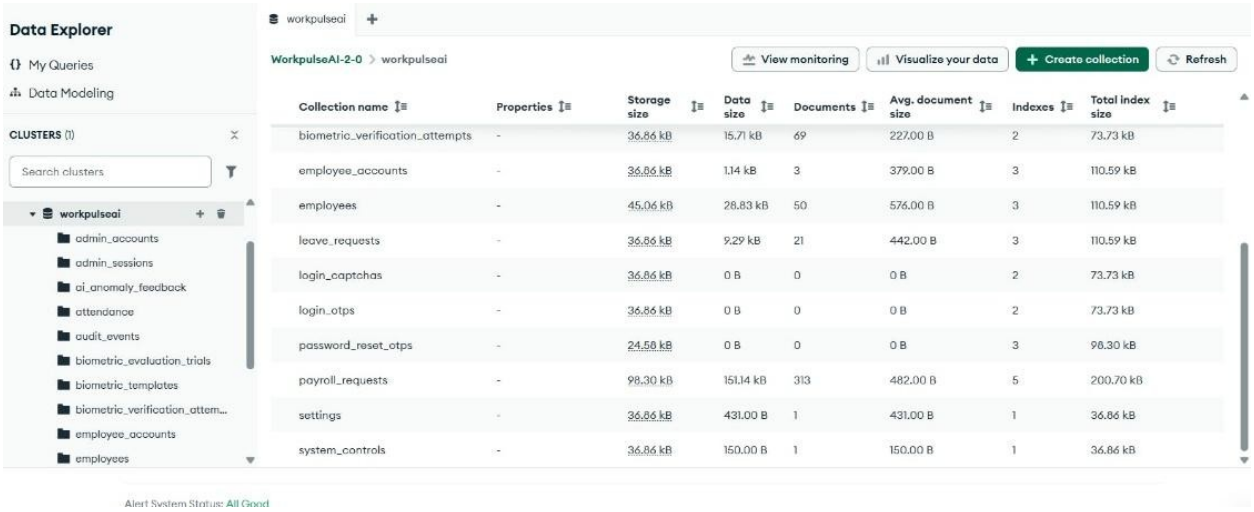


Figure 12. Database interface supplied in the manuscript

Personal Employee Information

The portal shows personal profile and attendance-related information. Protected changes remain subject to authorization, and administrators can notify the employee about relevant changes.

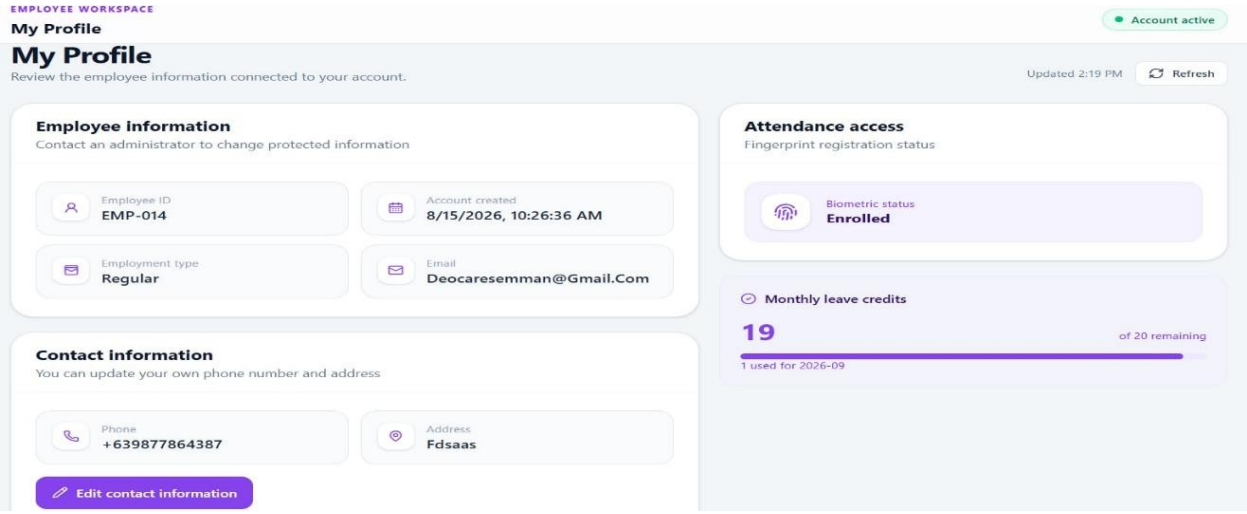


Figure 13. Employee profile interface supplied in the manuscript

Cash Advance Management

Employees request advances and accept the applicable repayment terms. Administrators review requests and separately confirm cash given. The system tracks outstanding debt and recorded repayments. The tests distinguish debt from an unfulfilled request and prevent carried source statements from counting the same repayment twice.

Company and System Settings

System Settings provide hourly-rate controls for Regular, Extra, and Manager / Supervisor. The combined management classification uses one company rate. Legacy supervisor records use the configured management rate, while completed attendance sessions retain their recorded rates. This preserves historical earnings when company defaults change.

Offline Saving and Synchronization

Supported requests are stored locally as Pending without being presented as approved or settled. Reconnection triggers session checks and employee request synchronization. Attendance waits for administrator password confirmation. The server rechecks current business rules and stable operation identifiers before marking an accepted action Synced. Failed validation produces Needs Review, and network or storage failures produce error feedback rather than false success.

Functional Verification Results

The ten technical checklist cases in the revised source Chapter 3 were evaluated using controlled local tests. The October 12, 2026 backend run completed with 204 passing checks, zero failures, and zero skipped checks; the production build and isolated Edge browser checks also passed. Backend checks use simulated records. Browser checks use actual IndexedDB and service worker functionality with simulated API responses. The Passed status refers to these controlled testing environments, not verification of the live deployed website. The rows group tested behaviors and are not participant trials.

Table 3. Original technical checklist cases and local verification results

No.	Test scenario	Actual result	Status
1	Future attendance time	Future times rejected by attendance validation; browser rejected a future capture without saving it	Passed
2	Overlapping sessions	Overlapping sessions rejected and existing attendance preserved	Passed
3	Settled payroll attendance edit	Changes to attendance covered by settled payroll rejected; original records retained	Passed
4	Unauthorized record access	An employee requesting E2 data received only their own E1 attendance, payroll, and requests; administrator API access denied	Passed
5	Offline queue and reconnection	Request persisted as Pending in IndexedDB, survived offline reload, and synchronized after reconnect; server validation tested separately	Passed
6	Duplicate synchronization	Retry reused the original operation ID; no duplicate business mutation, notification, or payroll recalculation in tested cases	Passed

No.	Test scenario	Actual result	Status
7	Network interruption	Simulated lost response retained the operation locally as queued; local-save feedback shown and retry did not duplicate the action	Passed
8	Local storage failure	Injected IndexedDB write failure surfaced an error and rejected the local save	Passed
9	Payroll carried balance	Prior unpaid amounts included in carried earnings and destination payout; concurrent creation and source-linking rules passed	Passed
10	Cash advance repayment	Preparation/approval did not reduce debt; confirmed settlement applied eligible deductions; carried sources did not duplicate repayment	Passed

In the controlled cash advance case, three eligible earnings periods produced a ₱600 repayment on settlement, reducing a ₱1,000 debt to ₱400. A second payroll case combined two ₱800 earnings periods with two ₱200 installments, producing ₱1,600 earnings, ₱400 repayment, and a ₱1,200 payout. These values are test inputs and outcomes, not actual employee transactions.

Security and Authentication Regression Verification

Following security changes, local verification on October 12, 2026 completed with 204 passing backend checks, zero failures, and zero skipped checks. The frontend production build and isolated Edge browser checks also passed. Authentication checks exercised the actual Express routes over local HTTP with simulated database records and intercepted email-provider responses. They verify the tested software behavior, not actual email delivery, live Atlas permissions, or live Vercel/Render behavior. The following table groups the necessary security regression scenarios; its rows are not participant trials.

Table 4. Security regression scenarios and observed outcomes

No.	Test scenario	Actual result	Status
1	Valid password and CAPTCHA followed by login OTP	Login generated a hashed OTP and mocked email request; a valid OTP created a session with HttpOnly and Secure cookie flags	Passed
2	Incorrect password/CAPTCHA or wrong/expired login OTP	Invalid credentials and codes were rejected without creating a session	Passed

No.	Test scenario	Actual result	Status
3	Simultaneous or repeated valid OTP submission	Only one concurrent request created a session; the other request and later replay were rejected	Passed
4	Password recovery and first-login password change	Recovery code/token replay rejected; password changed and old employee sessions revoked; first-login change enabled authorized employee API access	Passed
5	MongoDB operator objects or oversized identifiers in authentication; operator employee ID in payroll	Malformed authentication identifiers rejected before query access; payroll rejected an operator object as the employee ID	Passed
6	Anonymous API access, employee access to another employee or administrator functions, and cookie writes without CSRF	Unauthorized calls rejected; attendance, payroll, requests, and notifications restricted to the signed-in employee; missing CSRF rejected and logout invalidated the session	Passed
7	Repeated fingerprint requests and email rotation to evade a rate budget	The 31st fingerprint request returned HTTP 429; changing email did not reset the tested IP budget; retry after window expiry succeeded	Passed
8	Encrypted fingerprint template round-trip and altered ciphertext	Original test templates decrypted correctly; tampered ciphertext was rejected without physical scanner testing	Passed
9	Excel export after dependency updates	Generated workbook reopened successfully and retained the test payroll value	Passed

Dependency advisory checks reported zero known production-dependency vulnerabilities in both packages after the updates. This result is not a guarantee of security. Rate limits are process-local, and production proxy behavior requires verification. Atlas network restrictions, database-user permissions, live secrets, real backup restoration, and real OTP delivery remain deployment checks. Existing tester-account behavior was retained. Browser owner-identifier checks do not establish independent login sessions across tabs.

Usability Evaluation Using SUS

SUS was the only questionnaire administered for respondent evaluation. The computed overall score was 80.83 out of 100. This result describes perceived usability and is separate from functional correctness, fingerprint identification accuracy, or measured processing speed.

Table 5. SUS usability result

Measure	Result
Overall SUS score	80.83 out of 100
Adjective rating	Good
Acceptability	Acceptable

The result indicates favorable usability feedback under the study's interpretation. SUS is not a percentage of satisfied participants and does not provide separate security, performance-efficiency, or reliability ratings. Those additional quality questionnaires were not administered; therefore, no separate category means are reported.

Participant Profile

Table 6. Study population distribution

Classification	Number	Percentage
Owner	2	6.67%
Manager / Supervisor	2	6.67%
Regular employee	10	33.33%
Extra employee	16	53.33%
Total	30	100.00%

The distribution describes the study population supplied in the revised manuscript. Extra employees constitute the largest category. It does not represent response frequencies for interview topics, nor does it establish that every participant was interviewed.

Biometric and Deployment Evaluation Limitations

The available evaluation records do not establish physical fingerprint trial counts, failed captures, false identifications, or mean verification times. Consequently, numerical biometric accuracy and speed are not reported. The application tests verify rules surrounding attendance but do not substitute for controlled scanner trials.

The local browser checks used simulated API responses, so they do not establish live Vercel/Render outage behavior or production database recovery. The 204 passing backend checks support the specific tested conditions rather than an assertion that all possible errors have been eliminated. Analytical accuracy and long-term operational availability require

additional evaluation.

Chapter 4

CONCLUSIONS AND RECOMMENDATIONS

This chapter draws conclusions from the qualitative interview topics, the SUS usability result, and documented functional verification. These evidence sources address different aspects of WorkPulse and are interpreted within their respective limits.

Conclusions

1. The interview findings described workplace concerns involving attendance record accuracy, proxy-attendance risk, monitoring of work hours, payroll processing, and employee access to records. These topics informed the requirements for centralized workforce information. No separate workplace-problems survey was used to assign percentages to those concerns.

2. WorkPulse implements employee records, local fingerprint attendance, authorized manual attendance and corrections, notifications, leave and cash advance requests, payroll summaries, descriptive reports, and selected analytical functions. Administrator and employee access groups are maintained separately from the Regular, Extra, and Manager / Supervisor employee classifications.

3. The researchers' computed SUS score of 80.83 was interpreted as Good and Acceptable. This result supports favorable perceived usability. It does not independently establish technical correctness, security assurance, or fingerprint performance.

4. Local verification on October 11, 2026 reported 175 passing backend checks with zero failures. The production build and isolated browser checks completed successfully. Tested outcomes included attendance validation, payroll repayment handling, notification targeting, legacy role compatibility, local request storage, reconnect synchronization, account isolation, and duplicate prevention.

5. Cash advance requests and approval alone do not create debt. Confirmed disbursements establish outstanding balances, while posted repayments reduce them. Controlled tests showed that eligible installments accumulate across carried earnings periods and that carried source statements do not duplicate repayments.

6. Limited offline saving supports selected attendance actions and employee requests on prepared browsers with valid sessions. Server acceptance remains subject to authorization and current business rules. Fresh login, fingerprint transactions, approvals, and settlement actions

still require online services.

7. Fingerprint attendance depends on the configured local Windows reader and matching environment. The available records do not establish numerical biometric accuracy or processing speed. Live deployment recovery and forecasting accuracy also remain outside the evidence established by these local software checks.

Recommendations

1. Conduct controlled physical-reader trials using consenting participants. Record enrolled templates, genuine and unrecognized-finger trials, failed captures, identification outcomes, reader conditions, and elapsed verification times.

2. Maintain documented recovery procedures for the local workstation, reader software, and matching components. Obtain organizational approval for a supervisor-verified paper attendance log and its later reconciliation through authorized manual entry.

3. Verify the final Vercel/Render deployment using real session, ownership, notification, payroll, and outage scenarios. Evaluate the local-to-cloud attendance architecture before claiming that the cloud backend can use locally installed matching components.

4. Extend outage and recovery testing across browsers, devices, session expiry, prolonged disconnection, storage limits, duplicate retries, and conflicting updates. Retain test records for the environment and application version evaluated.

5. Improve attendance exception review, cash advance reconciliation, carried-balance reporting, and notification preferences beyond the existing functions.

6. Evaluate analytical outputs against relevant real outcomes using documented methods and error measures. Forecasts and risk indicators should continue to support human review rather than automatic employment decisions.

7. Retain interview records, the original SUS instrument and scoring sheets, test logs, and deployment evidence as supporting project materials. Future revisions should repeat the applicable functional checks when business rules change.